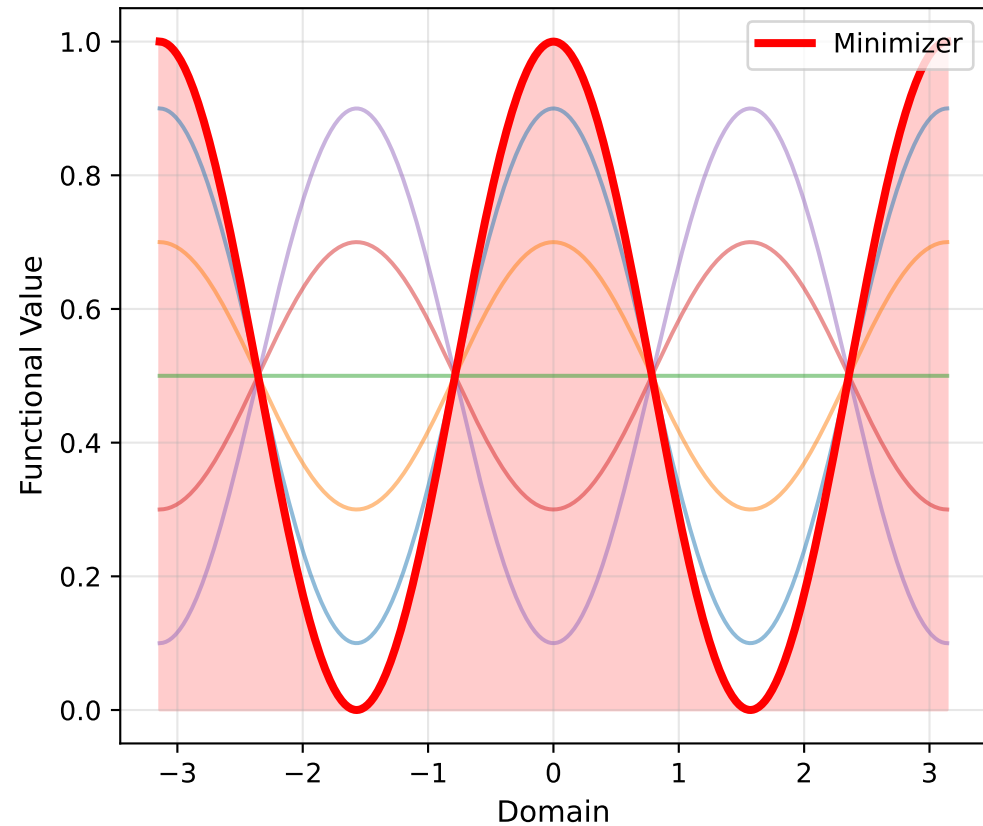


Variational Principle: Finding Extremals



Variational Problem:

$$\min_{u \in V} J(u) = \int_D F(u, \nabla u) dV$$

Leads to Euler Equation:

$$\frac{\partial F}{\partial u} - \nabla_i \frac{\partial F}{\partial u_i} = 0$$

Critical Point = Solution